SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

R. Geethika

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Max Marks: **50 Marks**

Date : _____

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at x = 6. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at x = 6.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 192425101

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	4
Task 4	4	4
Task 5	3	3
Task 6	3	2
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	4
Task 11	4	3
Task 12	8	7
TOTAL	**50**	45

Total in words:

Assignment - 2

MIA0204 - Fundamentals Of Machine learning

Name :- R. Geethika
Reg :- 192425101

Q1) Linear Regression Using Ordinary least Squares

45/50

Given

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

a) Fit the equation

$$y = w_0 + w_1 x$$

$n = 5$, $\Sigma x = 15$, $\Sigma y = 30.1$

$\Sigma x^2 = 55$, $\Sigma xy = 110.2$

$$w_1 = \frac{5(110.2) - 15(30.1)}{5(55) - 15^2}$$

$$= \frac{551 - 451.5}{275 - 225}$$

$$= \frac{99.5}{50} = 1.99$$

$$w_0 = \frac{30.1 - 15(1.99)}{5}$$

$$= \frac{0.25}{5} = 0.05$$

$$y = 0.05 + 1.99x$$

(b) Predict y at $x = 6$

$$y = 0.05 + 1.99(6)$$

$$= 0.05 + 11.94$$

$$y = 11.99$$

(c) Calculate Rss, MSE and R^2

x	Actual y	Predicted y	Residual
1	2.1	2.04	0.06
2	3.9	4.03	-0.13
3	6.2	6.02	0.18
4	7.8	8.01	-0.21
5	10.1	10.00	0.10

Residual sum of squares:

$$Rss = \sum (y - \hat{y})^2$$

$$= 0.06^2 + (-0.13)^2 + 0.18^2 + (-0.21)^2 + 0.10^2$$

$$= 0.0026 + 0.0169 + 0.0324 + 0.0441$$

$$Rss = 0.107$$

Mean squared error:

$$MSE = \frac{Rss}{n} = \frac{0.107}{5} = 0.0214$$

Coefficient of determination:

$$R^2 = 1 - \frac{RSS}{SST}$$

$$SST = 29.6756$$

$$R^2 = 1 - \frac{0.107}{29.6756}$$

$$\boxed{R^2 = 0.9973}$$

∴ The Model gives a very Good fit because R2 is close to 1.

Q2) Task - 2 - Ridge Regression

Ridge Regression uses L2 regularization to reduce Overfitting.

Its cost function is

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda \beta^2$$

Given the calculated values:-

- Original coefficient = 1.99
- Ridge Coefficient = 1.3267

Shrinkage

$$\frac{1.99 - 1.3267}{1.99} \times 100 = 33.33\%$$

∴ Ridge Regression Reduces the Coefficient by 33.33-1. Increasing λ reduces variance But increases Bias.

Q3) Task-3 - Naive Bayes classifier

Baye's theorem

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Given,

$P(spam) = 0.5$, $P(ham) = 0.5$

for offer = Yes and link = Yes:

- Without laplace smoothing

$P(spam/x) = 94.12\%$.

- With laplace smoothing

$P(spam|x) = 86.21\%$.

for offer = N and link = NO:

$P(spam|x) = 13.79\%$.

∴ Since the Probability of ham is higher classify the Message as ham.

Q4) Task 4 – logistic Regression

the sigmoid function is

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Initially:

$$P_1 = P_2 = 0.5$$

the calculated gradients are

$$dw_1 = 0.25,\ dw_2 = -0.25,\ db = 0$$

using learning rate $\alpha = 0.5$

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$w_2 = 0 - 0.5(-0.25) = 0.125$$

$$b = 0$$

Updated Probabilities

$$P_1 = 0.5312,\ P_2 = 0.4688$$

loss changes from 0.6931 and 0.6326.

∴ the loss decreases after Gradient descent, indicating improvement.

Q5) Task 5 – Decision Tree

Entropy formula:

$$H(S) = -\sum P_i \log_2 P_i$$

Calculated values:

- Parent entropy = 0.9710
- child Entropy1 = 0
- child entropy 2 = 0.9183
- weighted entropy = 0.5510.

Information gain:

$$IG = 0.9710 - 0.5510$$

$$IG = 0.4200$$

3

Gini impurity

$$Gini = 1 - \sum P_i^2$$

- Parent Gini = 0.48
- Weighted Gini = 0.2667
- Gini decrease =

$$0.48 - 0.2667$$

$$= 0.2133.$$

∴ the split reduces impurity and is useful for the decision tree.

Q1) Task 6- Cost Complexity Purning

The Pruning Criterion is

$$R_\alpha(T) = R(T) + \alpha|T|$$

Critical alpha:

$$\alpha = \frac{R(T) - R(T')}{|T'| - |T|}$$

$\alpha = 0.01667$

Comparision

- When $\alpha = 0.01$, α is smaller than the Critical value, keep the subtree.
- When $\alpha = 0.02$, α is Greater than the Critical value. Prune the subtree

∴ Higher α Produces a simpler Decision tree.

Q7) Task 7 – Neural Network Backpropagation

the sigmoid activation function is

$$f(x) = \frac{1}{1+e^{-x}}$$

Calculated formula Propagation

- Hidden Output = 0.7311
- Predicted Output = 0.5904.

Initial Error

$$E = \frac{1}{2}(\text{target} - \text{Output})$$

$$E = 0.0829$$

After Backpropagation, updated weights are

$$V = 0.5369$$

$$W_1 = 0.4049, \quad W_2 = 0.3097$$

New error:

$$E \approx 0.0811.$$

∴ The error decreases after updating the weights.

Q2) Task 2 – Support Vector Machine (SVM)

The decision Boundary is

$$W^T x + b = 0$$

Given,

$$W = (0.5, 0.5), \quad b = -2$$

Decision function:

$$f(x) = 0.5x_1 + 0.5x_2 - 2$$

For (3,3):

$$f(x) = 1$$

class = +1.

For (1,1):

$$f(x) = -1$$

class = -1

Margin width:

$$\frac{2}{\|W\|} = \frac{2}{\sqrt{0.5^2 + 0.5^2}} = 2.8284$$

∴ The SVM separates the two classes using the decision Boundary.

9) Task 9 – Ensemble Majority Voting

For independent classifiers with Accuracy $p = 0.7$ Majority voting Accuracy is calculated using the Binomial Probability.

For 3 classifiers,

$$P = \binom{3}{2}(0.7)^2(0.3) + (0.7)^3$$

$$\boxed{P = 78.4\%}$$

For 5 classifiers

$$P = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

$$\boxed{P = 83.69\%}$$

∴ Five classifiers gives Better clarity Majority Voting Accuracy than three.

Q10) Task 10 – Ada Boost

Weighted Error

$$\epsilon = 0.4$$

classifier Weight

$$\alpha = \frac{1}{2} \ln\left(\frac{1-e}{e}\right)$$

$$= \frac{1}{2} \ln\left(\frac{0.6}{0.4}\right)$$

$$\boxed{\alpha = 0.2027}$$

After Updating and Normalizing the sample Weights.

- Correctly classified sample Weight = 0.1667,
- Incorrectly classified sample Weight = 0.25

∴ Ada Boost gives More Weight to incorrectly classified sample Weight so that the next learner focuses on them.

Q11) Task 11 – Linear Discriminant Analysis (LDA)

The LDA Direction is

$$W = \Sigma_W^{-1} (M_1 - M_2)$$

Calculated Direction

$$W = (10/7, -12/7.)$$

Equal Prior threshold

$$t = 0.2857$$

for the given point (2.5, 1)

$$y = w^T x$$

$$= 10/7(2.5) - 12/7(1)$$

$$= 1.2857$$

3

Since the projected value is above the threshold classify it as class w_2.

For Unequal Priors, the threshold is 1.6720 the point is still classified as class w_2.

Q12) Task 12 – classifier Evaluation

Important Formulas

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

Calculated Results

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
Precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-Score	53.85%	46.58%	83.16%

Cost Comparision

- Model A = ₹ 19.5 lakh
- Model B = ₹ 16.5 lakh
- Model C = ₹ 21.5 lakh

∴ Model B has the lowest Cost and highest recall. Model C has the highest Accuracy and F1-Score. The Best Model depends on whether cost, recall, Or Overall Accuracy is the Priority.